Supplemental Tables

Is emotional processing fine-tuned to speaker gender, identity and phonetic content? - Evidence from simultaneous opposite adaptation

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Table S1

Summary of the acoustic characteristics of the stimuli used as adaptors

	female speakers		male speakers	
	anger	**fear**	**anger**	**fear**
F0 Mean	366	288	264	205
F0 SD	67	19	46	27
F0 Intonation	259	79	171	95
F0 Glide	-60	3	-41	12
Alpha Ratio	1.0	2.3	1.1	2.3
HNR	12.8	21.3	7.3	19
Jitter	0.004	0.004	0.011	0.004
Shimmer	0.035	0.013	0.063	0.013
Duration	885	760	731	854

Note. All acoustic parameters were adapted from McAleer et al. (2014) and extracted using Praat software (Boersma, 2018). For F0 extraction, pitch ranges were set to 170 - 600 Hz for female and 100 - 370 Hz for male stimuli. F0 Intonation = F0max – F0min; F0 Glide = F0End – F0Start; Alpha ratio (a measure of the spectral slope) = ratio of mean energy within low (0–1 kHz) and high frequencies (1–5 kHz), computed from the long-term average spectrum; HNR (harmonics-to-noise ratio) was extracted with the cross-correlation method (mean value; time step = 0.01 s; min pitch = 75 Hz; silence threshold = 0.1, periods per window = 1.0), Jitter as a measure of local F0 variation (shortest period = 0.0001 s; longest period = 0.02 s; max. period factor = 1.3); Shimmer as a measure of local amplitude variation (shortest period = 0.0001 s; longest period = 0.02 s; max. period factor = 1.3; max. amplitude factor = 1.6); Duration in ms.

Table S2

Summary of the acoustic characteristics of target stimuli – Part 1

SpGender	tML	*F0 Mean*	*F0 SD*	*F0 Intonation*	*F0 Glide*
f	20	298	21	90	-20
f	30	304	25	102	-27
f	40	311	30	118	-34
f	50	318	36	139	-40
f	60	325	42	165	-50
f	70	334	49	186	-55
f	80	343	55	208	-54
m	20	212	27	98	-8
m	30	216	29	104	-18
m	40	221	30	110	-21
m	50	226	33	119	-29
m	60	232	36	128	-35
m	70	238	38	138	-38
m	80	244	42	150	-45

Note. All acoustic parameters were extracted like the ones reported in Table S1. SpGender = speaker gender (f = female, m = male), tML = target morph level, corresponding to the % of anger in the voice.

Table S3

Summary of the acoustic characteristics of target stimuli – Part 2

SpGender	tML	*Alpha Ratio*	*HNR*	*Jitter*	*Shimmer*	*Duration*
f	20	1.9	20.8	0.003	0.009	785
f	30	1.7	19.8	0.003	0.011	798
f	40	1.6	18.8	0.003	0.013	812
f	50	1.4	17.9	0.003	0.017	825
f	60	1.3	17.0	0.003	0.022	833
f	70	1.2	16.1	0.003	0.024	845
f	80	1.2	15.2	0.003	0.026	860
m	20	1.9	17.5	0.004	0.014	831
m	30	1.7	16.1	0.004	0.018	816
m	40	1.6	14.7	0.004	0.024	804
m	50	1.5	13.2	0.005	0.029	791
m	60	1.4	11.9	0.005	0.034	776
m	70	1.3	10.7	0.006	0.042	767
m	80	1.2	9.7	0.006	0.045	756

Note. All acoustic parameters were extracted like the ones reported in Table S1. SpGender = speaker gender (f = female, m = male), tML = target morph level, corresponding to the % of anger in the voice.

Table S*4*

Full Parameter List of Multilevel Logistic Regression for Experiment 3

Fixed effects	***Estimate***	***SE***	***z***	***p***
(Intercept)	0.32	0.32	1.01	0.311
tML	**1.65**	**0.09**	**18.15**	**<0.001**
PW: PW2*	**-1.64**	**0.12**	**-13.75**	**<0.001**
PW: PW3	**-1.05**	**0.12**	**-9.03**	**<0.001**
PW: PW4	-0.41	0.11	-3.55	0.000
AdaptCond: 2**	-0.23	0.11	-2.13	0.033
tML x PW: W2	-0.03	0.13	-0.25	0.805
tML x PW: W3	**0.41**	**0.14**	**2.92**	**0.003**
tML x PW: W4	**0.68**	**0.15**	**4.63**	**<0.001**
tML x AdaptCond: 2	0.12	0.13	0.95	0.343
PW: W2 x AdaptCond: 2	0.10	0.17	0.57	0.568
PW: W3 x AdaptCond: 2	0.09	0.17	0.53	0.595
PW: W4 x AdaptCond: 2	0.13	0.16	0.81	0.417
tML x PW: W2 x AdaptCond: 2	-0.16	0.19	-0.84	0.404
tML x PW: W3 x AdaptCond: 2	-0.02	0.20	-0.11	0.913
tML x PW: W4 x AdaptCond: 2	**-0.57**	**0.20**	**-2.84**	**0.005**

*Note. The response is the classification of stimuli as angry or fearful. The target morph level (tML) was z-standardized for model estimation. *PW1: /belam/, PW2: /molen/, PW3: /namil/ and PW 4: /loman/. ** AdaptCond1 angry-PW1|2 and fearful-PW3|4; AdaptCond2: angry-PW3|4 and fearful-PW1|2.*

Table S5

Additional Multilevel Logistic Regressions for Experiment 3, separate for males and females

Model Exp 3 - female stimuli only

*Response ~ tML * Pseudoword * AdaptCond + (1|SpID) + (1 | Participant)*

Predictor	***Chi2***	***p***
Morph Level (tML)	**1998.34**	**<.001**
Pseudoword (PW)	**372.49**	**<.001**
AdaptCond	2.36	0.124
tML x PW	**9.76**	**<.001**
tML x AdaptCon	0.60	0.438
AdaptCond x PW	0.80	0.850
tML x PW x AdaptCond	**10.74**	**0.013**

Fixed effects	***Estimate***	***SE***	***z***	***p***
(Intercept)	0.35	0.14	2.4	**0.015**
tML	1.60	0.13	12.5	**<0.001**
PW: PW2*	-2.04	0.19	-11.0	**<0.001**
PW: PW3	-1.04	0.16	-6.6	**<0.001**
tML x PW: W4 x AdaptCond: 2	-0.55	0.26	-2.12	**0.034**

Model information		
AIC/BIC	4111	4227
N observations	4686	
N participants	42	
N Speaker Identity	2	

Random effects	***Var***	***SD***
Participant - Intercept	0.39	0.62
Speaker Identity (SpID) - Intercept	<0.01	0.02

Model Exp 3 - male stimuli only

*Response ~ tML * Pseudoword * AdaptCond + (1|SpID) + (1 | Participant)*

Predictor	***Chi2***	***p***
Morph Level (tML)	**2231.06**	**<.001**
Pseudoword (PW)	**136.73**	**<.001**
AdaptCond	**6.58**	**0.010**
tML x PW	**83.62**	**<.001**
tML x AdaptCon	2.67	0.102
AdaptCond x PW	1.28	0.733
tML x PW x AdaptCond	3.71	0.295

Fixed effects	***Estimate***	***SE***	***z***	***p***
(Intercept)	0.30	0.65	0.46	0.646712
tML	1.80	0.13	13.47	**<0.001**
PW: PW2*	-1.39	0.17	-8.41	**<0.001**
PW: PW3	-1.11	0.17	-6.36	**<0.001**
PW: PW4	-0.67	0.18	-3.74	**<0.001**
AdaptCond: 2**	-0.37	0.16	-2.29	**0.022**
tML x PW: W3	0.65	0.21	3.05	**0.002**
tML x PW: W4	1.23	0.24	5.08	**<0.001**

Model information		
AIC/BIC	3688	3804
N observations	4685	
N participants	42	
N Speaker Identity	2	

Random effects	***Var***	***SD***
Participant - Intercept	0.76	0.87
Speaker Identity (SpID) - Intercept	0.79	0.89

*Note. The response is the classification of stimuli as angry or fearful. The target morph level (tML) was z-standardized for model estimation. This table shows only a subset of effects for readability. *PW1: /belam/, PW2: /molen/, PW3: /namil/ and PW 4: /loman/. ** AdaptCond1 angry-PW1|2 and fearful-PW3|4; AdaptCond2: angry-PW3|4 and fearful-PW1|2.*

References

Boersma, P. (2018). Praat: Doing phonetics by computer [Computer program]: Version 6.0.46, retrieved January 2020 from http://www.praat.org/. *Http://www. Praat. Org*.

McAleer, P., Todorov, A., & Belin, P. (2014). How do you say 'Hello'? Personality impressions from brief novel voices. *PLoS One*, *9*(3), e90779. https://doi.org/10.1371/journal.pone.0090779